Feature Engineering and Baseline Model

Feature engineering was performed on the Telco Customer Churn dataset by creating `tenure_bucket`, `avg_monthly_spend`, and `services_count`. Categorical variables were encoded using One-Hot Encoding, while numerical features were imputed and standardized.

A Logistic Regression model was trained using class-weight balancing to address the imbalance between churn and non-churn customers.

Baseline Model Results:

- Accuracy: 75.30%
- Precision: 52.40%
- Recall: 75.94%
- F1-score: 62.01%
- ROC-AUC: 0.8401

The baseline model achieved a ROC-AUC of 0.8401 and a recall of 75.94%, indicating useful churn-detection capability. Precision can be improved in future iterations to reduce unnecessary customer-retention offers.